

Subspace

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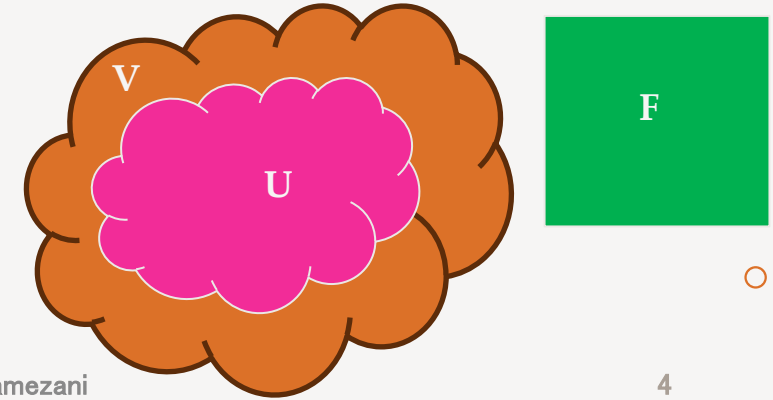
Subspace Definition

Subspace

- Zero vector is a subspace of every vector space.
- Vector space is a subspace of itself.

Definition

A **non-empty subset** of vector space for which closure holds for addition and scalar multiplication is called a subspace.
Subspace: If V is a vector space and subset $U \subseteq V$, then U is itself a vector space with the **same addition and scalar multiplication as V**.



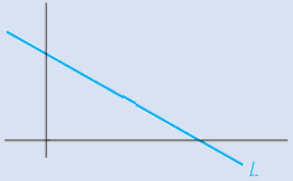
Subspace

A subspace of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has these properties:

- **The zero vector is in H.**
- **For each u and v in H, the sum u + v is in H.**
- **For each u in H and each scalar c, the vector cu is in H.**

Example

- Is L subspace of $\mathbb{R}^2$?



- The vector space $\mathbb{R}^2$ is a subspace of $\mathbb{R}^3$?
- Is H a subspace of $\mathbb{R}^3$? $H = \{ \begin{bmatrix} s \\ t \\ 0 \end{bmatrix} : s \text{ and } t \text{ are real} \}$

Subspace

Theorem (1)

A non-empty subset U of V is a subspace of V if and only if for each pair of vectors b, c in U and each scalar α in F the vector $\alpha b + c$ is again in U.

Proof:

Subspace

Example

- Every vector space with more than one non-zero member has at least _____ subspaces.
- Name subspace for $\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^4$
- Let H be the set of all vectors of the form $\begin{bmatrix} a-3b \\ b-a \\ a \\ b \end{bmatrix}$ where a,b are arbitrary scalars. That is, let$H = \left\{ \begin{bmatrix} a-3b \\ b-a \\ a \\ b \end{bmatrix} : a, b \ in \ R \right\}$. Show that H is a subspace of $\mathbb{R}^4$.

02

Intersection of subspaces

Intersection of subspaces

Theorem (2)

If W_1 and W_2 are subspaces of V, then $W_1 \cap W_2$ is a subspace.

Proof:

Theorem (3)

Intersection of any collection of subspaces of a vector space V, is a subspace of V.

Proof:

03

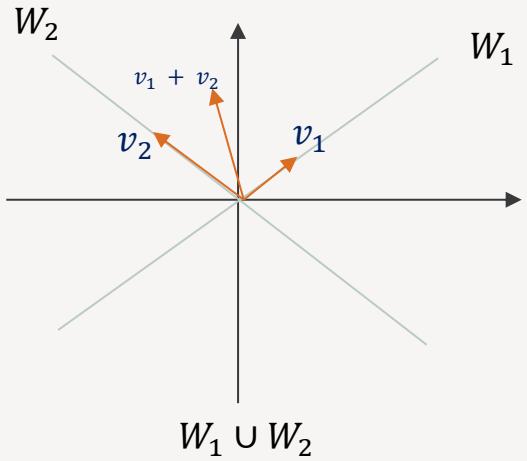
Union of subspaces

Union

Theorem (4)

The union of two sub-spaces may not be a subspace.

Proof:



Union

Theorem (5)

Fact: The union of two sub-spaces is not a subspace unless one is contained in the other.

W_1 and W_2 are subspaces of V, then $W_1 \cup W_2$ is subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$

Proof:

04

Span and Subspace

Span or linear hull

Definition

If $v_1, v_2, v_3, \dots, v_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $v_1, v_2, \dots, v_p$ is denoted by Span $\{v_1, v_2, \dots, v_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned (or generated)** by $v_1, v_2, \dots, v_p$.

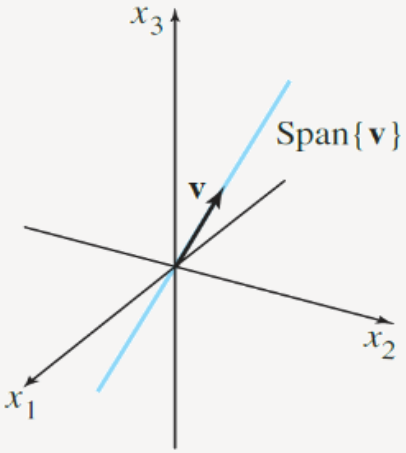
That is, Span$\{v_1, v_2, \dots, v_p\}$ is the collection of all vectors that can be written in the form:

$$c_1 v_1 + c_2 v_2 + \dots + c_p v_p$$

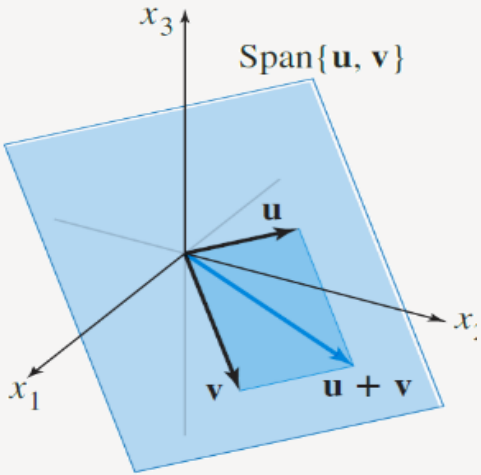
with $c_1, c_2, \dots, c_p$ being scalars.

Span Geometry

v and u are non-zero vectors in $\mathbb{R}^3$ where v is not a multiple of u



Span {v} as a
line through the origin.



Span {u, v} as a
plane through the origin.

Span and Subspace

Theorem (6)

If $v_1, v_2, \dots, v_p$ are in a vector space V, then Span $\{v_1, v_2, \dots, v_p\}$ is a subspace of V.

Proof:

05

Sum of subspaces

Linear Sum of subspaces

Definition

Let A and B be non-empty subsets of a vector space V. The sum of A and B, denoted A+B, is the set of all possible sums of elements from both subsets:

$$A + B = \{a + b : a \in A, b \in B\}$$

Example

- $A = \{t_1(2,3) | t_1 is\ scalar\}\ B = \{t_2(3,1) | t_2\ is\ scalar\}$, A+B?

- $A = \{t_1(1,2,0) | t_1 is\ scalar\}\ B = \{t_2(0,1,2) | t_2\ is\ scalar\}$, A+B?

Linear Sum of subspaces

Theorem (7)

If $W_1, \dots, W_m$ are subspaces of V, then $W_1 + \cdots + W_m$ is a subspace of V.

06

Direct sum of subspaces

Direct sum

Definition

U + W is called a **direct sum**, if any element in U + W can be written uniquely as u + w where $u \in U \; and \; w \in W$ (Notation: $U \oplus W$)

Example

Check where sum of following elements is a direct sum?

a) $\mathrm{U} = \begin{bmatrix} a \\ b \\ 0 \end{bmatrix}, \mathrm{W} = \begin{bmatrix} 0 \\ 0 \\ c \end{bmatrix}$

b) $\mathrm{U} = \begin{bmatrix} a \\ b \\ 0 \end{bmatrix}, \mathrm{W} = \begin{bmatrix} 0 \\ c \\ d \end{bmatrix}$

Direct Sum

Theorem (8)

If U and W are subspaces of V, then the sum is a direct sum $U \oplus W$, if and only if

$$U \cap W = \{0\}$$

Proof:

Direct Sum

Example

Let E denote the set of all polynomials of even powers.
$E = \{a_n t^{2n} + a_{n-1} t^{2n-2} + \ \ldots + a_0\}$, and O be the set of all polynomials of odd powers :
$O = \{a_n t^{2n+1} + a_{n-1} t^{2n-1} + \ \ldots + a_1 t\}$.
The set of all polynomials P is a direct sum of E and O :

$$P = E \oplus O$$

It is easy to see that any polynomial (or function) can be uniquely decomposed into direct sum of its even and odd counterparts:

$$p(t) = \frac{p(t) + p(-t)}{2} + \frac{p(t) - p(-t)}{2}$$

Subspace Example

Example

Prove set of all bound functions such as

$$W = \{f(x) \mid \exists M \in R \; such \; that \; |f(x)| \leq M, \forall x \in R\}$$

is a subspace of $V = \{all \; functions \; from \; R \; to \; R\}$

Note

Triangle Inequality for Real Numbers

$$|a + b| \leq |a| + |b|$$

Resources

- LINEAR ALGEBRA: Theory, Intuition, Code, David Cherney.
- Chapter 4 of Elementary Linear Algebra with Applications
- Chapter 3 of Applied Linear Algebra and Matrix Analysis